

Answer all the following questions:

- I. Write the DTD file for the following XML document, and add the DOCTYPE tag into this XML document: **(6 pts)**

```
<?xml version = "1.0" encoding= "UTF-8"?>
<personal_files>
  <student id = "s1" lang= "En">
    <firstName>Samir</First Name>
    <lastName>Alzaime</last Name>
    <address> Beirut</address>
    <phone>01 /304050 </phone>
    <mobile> 70 / 657865</mobile>
    <date_Of_berth> 30/6/2000 </date_Of_berth>
  </student>
  <student id= "s2">
    <prenome> Lila </prenome>
    <nome> Karam </nome>
    <adress> zalka </adress>
    <telephone> 04 / 658487</telephone>
    <cellular> 76 / 976457 </cellular>
    <date_de_naissance> 30/9/2001 </date_de_naissance>
  </student>
  ...
  ...
</personal_files>
```

Note : when we start by a language for a student we must continue with this language.

- II. For this XML document write the corresponding Schema file, and add the link and name space attributes to this XML document: **(6 pts)**

```
<?xml version= "1.0" encoding= "UTF-16"?>
<schedules>
  <course lang= "FR" code= "X1" >
    <titre> XML </titre>
    <jour> lundi </jour>
    <professeur > Nabil Baze</professeur>
    <temps> 2 H @ 10 AM</temps>
    <sall> B101< /sall>
  </course>

  <course lang= "En" code= "UML1" >
    <title> UML </title>
    <day> Tuesday </day>
    <teacher > Jamal Azez</teacher>
    <time> 2 H @ 10 AM</time>
    <hall>A102</hall>
  </course>
  <course ...> </course>
  <course ...> </course>
</schedules>
```

Note : when we start by a language for a course we must continue with this language.

III. In the exercise number 2 there is a schedule, for an institute, distribute like XML document, write an XSL file to transform its content into two tables first for French language and the second for English language. Don't forget to name the table's columns like the name of tags. **(5pts)**

IV. Having the following XML document: **(3 pts)**

```

1      <?xml version= "1.0" ?>
2      <person_address>
3      <address type= "Lebanon-address ">
4      <name> Samir Alhoss </name>
5      <street> Alhamra </street>
6      <city> Beirut </city>
7      < building > Alariss</building>
8      <phone> 01765495</phone>
9      </address>

10     <address type= "Lebanon-address ">
11     <name> Salem Alkodarji </name>
12     <street> haret hrake </street>
13     <city> Hara </city>
14     < building > Alkasfe</building>
15     <phone> 01566899</phone>
16     </address>

17     <address type= "Lebanon-address ">
18     <name> Jamila Aldibce </name>
19     <street> Aine alromana </street>
20     <city> Alay </city>
21     <building> Aldibec</building>
22     <phone> 05575757</phone>
23     </address>
24     </person_address>

```

What are the results of the XPATH expression applied to the above XML document? Indicate the number of line or lines in your answer:

- ❖ //address/name
- ❖ //*@
- ❖ //*start-with(name(), "p")
- ❖ //*string-length()= "4"
- ❖ //*contains(name(), "y")
- ❖ //*last()